

## Trig 6

### Review

$$\begin{aligned} 1. \quad &= 1 + 30(2x)(-1) + 435(2x)^2(1) + 4060(2x)^3(-1) \\ &= 1 - 60x + 1740x^2 - 32480x^3 \end{aligned}$$

So the coefficient of  $x^2$  is 1740 and the coefficient of  $x^3$  is  $-32480$ .

2. direct proportion: A, E

inverse proportion: B, D, F

3. (C)  $y = 2 \cos x$

4.  $f(x)$  is increasing when  $f'(x) > 0$ .

$$\begin{aligned} f'(x) &= 3x^2 - 12x + 9 \\ &= 3(x - 3)(x - 1) \end{aligned}$$

5. Area:

$$\begin{aligned} A &= \frac{1}{2}r^2\theta \\ 2\pi &= \frac{\pi}{8}r^2 \\ r &= 4 \end{aligned}$$

Then the perimeter can be found by

$$\begin{aligned} &= 2r + r\theta \\ &= r(2 + \theta) \\ &= 4\left(2 + \frac{\pi}{4}\right) \\ &= 8 + \pi \\ &= 11.14 \end{aligned}$$

### Consolidation

$$\begin{aligned} 1. \quad &\sin x = \frac{3}{5} \\ &x = 0.6435011088 \\ &\tan x = \frac{3}{4} \\ &\cot x = (\tan x)^{-1} \\ &= \frac{4}{3} \end{aligned}$$

$$\begin{aligned} 2. \quad \text{i. } &x = \frac{3}{\pi}, \frac{3}{4\pi} \\ \text{ii. } &\cos x = \frac{1}{\sqrt{2}} \rightarrow x = \frac{\pi}{4}, \frac{7\pi}{4} \\ \text{iii. } &\sin x = -\frac{1}{2} \rightarrow \frac{7\pi}{6}, \frac{11\pi}{6} \end{aligned}$$

$$\begin{aligned}
3. \quad \text{i.} \quad \tan^2 \theta + \cot^2 \theta &= \frac{\sin^2 \theta}{\cos^2 \theta} + \frac{\cos^2 \theta}{\sin^2 \theta} \\
&= \frac{1 - \cos^2 \theta}{\cos^2 \theta} + \frac{1 - \sin^2 \theta}{\sin^2 \theta} \\
&= \sec^2 \theta - 1 + \csc^2 \theta - 1 \\
&= \sec^2 \theta + \csc^2 \theta - 2
\end{aligned}$$

$$\begin{aligned}
\text{ii.} \quad \sec^2 \theta &= \frac{1}{\cos^2 \theta} \\
&= \frac{1}{1 - \sin^2 \theta} \\
&= \frac{\frac{1}{\sin \theta}}{\left(\frac{1}{\sin \theta}\right) - \sin \theta} \\
&= \frac{\csc \theta}{\csc \theta - \sin \theta}
\end{aligned}$$

$$\begin{aligned}
\text{iii.} \quad (1 - \cos \theta)(1 + \sec \theta) &= 1 - \cos \theta + \sec \theta - 1 \\
&= \frac{1}{\cos \theta} - \cos \theta \\
&= \frac{1 - \cos^2 \theta}{\cos \theta} \\
&= (\sin^2 \theta) \cos \theta \\
&= \sin \theta \tan \theta
\end{aligned}$$

$$\begin{aligned}
4. \quad \text{i.} \quad \frac{2 \tan x}{\sin x} &= 3 \\
\frac{1}{\cos x} &= \frac{3}{2} \\
\cos x &= \frac{2}{3} \\
x &= 48.2^\circ, 311.8^\circ
\end{aligned}$$

$$\begin{aligned}
\text{ii.} \quad \tan x (\tan x - \sec x) &= 0 \\
\tan x \left( \frac{\sin x - 1}{\cos x} \right) &= 0 \\
\tan x = 0 &\rightarrow x = -180, 0, 180 \\
\frac{\sin x - 1}{\cos} x = 0 &\rightarrow \text{no solutions}
\end{aligned}$$

$$\begin{aligned}
5. \quad \text{i.} \quad 2 \tan x &= 1 + \frac{1}{\tan x} \\
2 \tan^2 x &= \tan x + 12 \tan^2 x - \tan x - 1 = 0 \\
\tan x = 1 &\rightarrow x = -45, 45 \\
\tan x = -\frac{1}{2} &\rightarrow x = -26.6, 26.6
\end{aligned}$$

$$\begin{aligned}
 \text{ii.} \quad & 2 \cos^2 x = 3 \cos x - 1 \\
 & 2 \cos^2 x - 3 \cos x + 1 = 0 \\
 & (2 \cos x - 1)(\cos x - 1) = 0 \\
 & \cos x = \frac{1}{2} \rightarrow x = 60, -60 \\
 & \cos x = 1 \rightarrow \text{no solutions (sec } x = 1 \text{ is undefined)}
 \end{aligned}$$

$$\begin{aligned}
 \text{iii.} \quad & \sin^2 x = \cos x + 5 \cos^2 x \\
 & 1 - \cos^2 x = \cos x + 5 \cos^2 x \\
 & 6 \cos^2 x + \cos x - 1 = 0 \\
 & (3 \cos x - 1)(2 \cos x + 1) = 0 \\
 & \cos x = \frac{1}{3} \rightarrow x = 70.5, 70.5 \\
 & \cos x = -\frac{1}{2} \rightarrow x = -120, 120
 \end{aligned}$$

### Exam Questions

$$\begin{aligned}
 1. \text{ a.} \quad & \sin \alpha = \frac{a}{c} \\
 & = \frac{2}{3}
 \end{aligned}$$

Also, from Pythagoras' Theorem,

$$\begin{aligned}
 a^2 + b^2 &= c^2 \\
 2^2 + b^2 &= 3^2 \\
 b &= \pm\sqrt{5}
 \end{aligned}$$

So

$$\begin{aligned}
 \cos \alpha &= \frac{b}{c} \\
 &= \frac{\sqrt{5}}{3} \text{ and } -\frac{\sqrt{5}}{3}.
 \end{aligned}$$

$$\begin{aligned}
 \text{b.} \quad & \frac{2 \sin^2 \beta - 7 \cos \beta}{\cos^2 \beta} + 5 = 0 \\
 & \frac{2 - 2 \cos^2 \beta - 7 \cos \beta}{\cos^2 \beta} + 5 = 0 \\
 & 2 + 3 \cos^2 \beta - 7 \cos \beta = 0 \\
 & (3 \cos \beta - 1)(\cos \beta - 2) = 0
 \end{aligned}$$

Then

$$\begin{aligned}
 \cos \beta &= \frac{1}{3} \rightarrow \sec \beta \rightarrow 3 \\
 \cos \beta &= 2 \rightarrow \text{no solutions}
 \end{aligned}$$

2. a.

$$\begin{aligned}\cos A + \sin A \tan A &= \cos A + \frac{\sin^2 A}{\cos A} \\ &= \frac{\cos^2 A + \sin^2 A}{\cos A} \\ &= \frac{1}{\cos A} = \sec A\end{aligned}$$

b.

$$\begin{aligned}\frac{2 \tan \theta}{1 - \tan^2 \theta} &= 3 \tan \theta \\ 2 \tan \theta &= 3 \tan \theta - 3 \tan^3 \theta \tan \theta (3 \tan^2 \theta - 1) = 0\end{aligned}$$

So

$$\begin{aligned}\tan \theta = 0 &\rightarrow \theta = 0, 180 \\ \theta = \frac{1}{\sqrt{3}} &\rightarrow \theta = 30\end{aligned}$$